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BUS7C2

Finance and Accounting for Business

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Lecture 9: Risk, Return and Cost of Capital

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Learning Objective

At the end of this lecture; students will:

1. **Critically assess** the interplay between risk, return, and capital costs, distinguishing key capital types and their financial implications.
2. **Apply** valuation models to calculate costs of equity and debt.
3. **Formulate** and defend WACC using market-based inputs, articulating core assumptions, calculation approaches, and contextual limitations.

Risk and Return

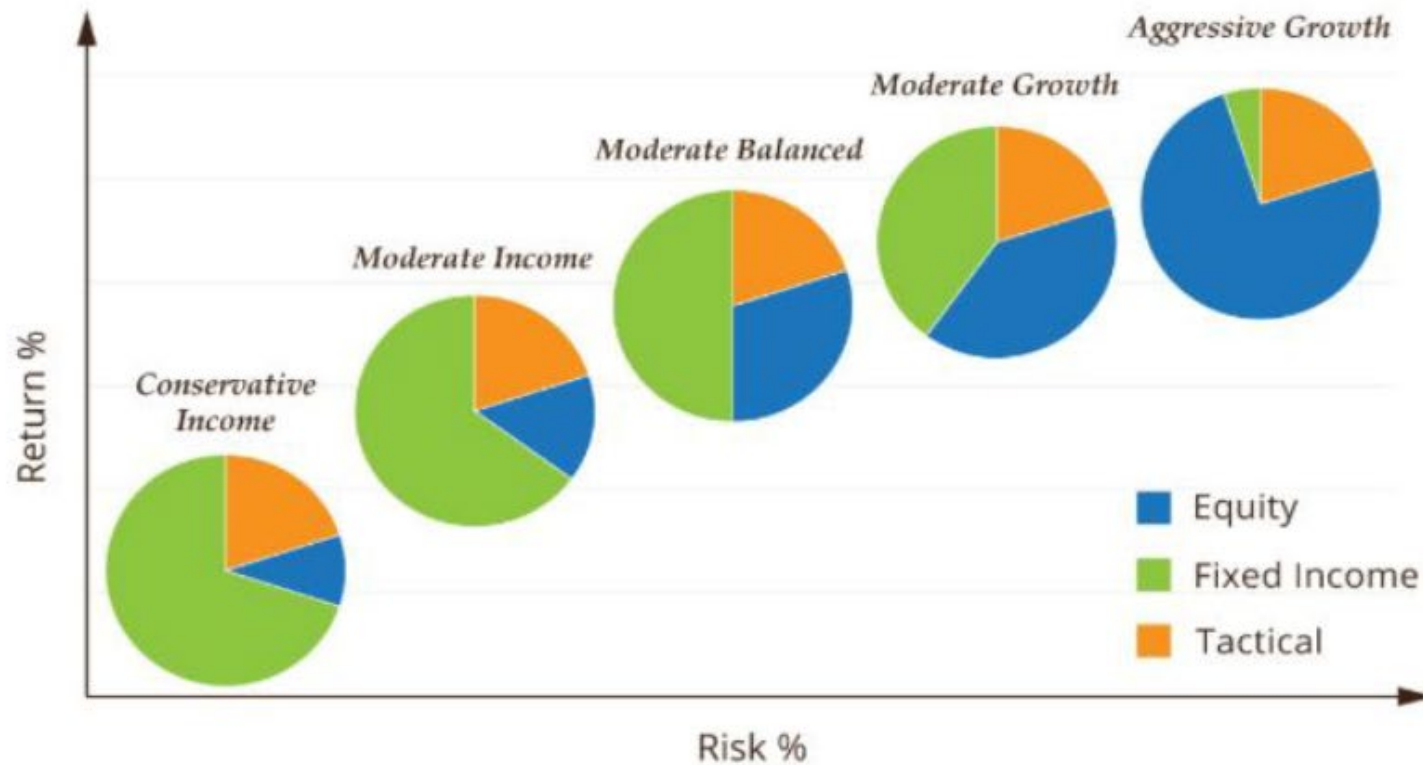
- The rate of return that investors require for an investment depends on the risk associated with that investment
 - The greater the risk, the larger the return investors require as compensation for bearing that risk
- The *rate of return* is what you earn on an investment, stated in percentage terms
- *Risk* is a measure of how certain you are that you will receive a particular return

Relation Between Risk and Return

- The greater the risk associated with an investment, the greater the return investors expect
- Investors want the highest return for a given level of risk or the lowest risk for a given level of return

Stock	Expected Return (%)	Risk Level (%)
A	12	12
B	12	16
C	16	16

Consider risk tolerance, investment objectives, time horizon



Holding Period Returns

- The total holding period return (R_T) consists of capital appreciation (R_{CA}) and income (R_I)

$$R_{CA} = \frac{\text{Capital Appreciation}}{\text{Initial Price}} = \frac{P_1 - P_0}{P_0} = \frac{\Delta P}{P_0}$$
$$R_I = \frac{\text{Cash Flow}}{\text{Initial Price}} = \frac{CF_1}{P_0}$$

Equation 7.1

$$R_T = R_{CA} + R_I = \frac{\Delta P}{P_0} + \frac{CF_1}{P_0} = \frac{\Delta P + CF_1}{P_0}$$

Total Holding Period Return Example

One year ago today, you purchased a share of Twitter, Inc., stock for \$32.05. Today it is worth \$44.00. What total return did you earn on this stock over the past year if Twitter paid no dividend? Since Twitter paid no dividend, and assuming you received no other income from holding the stock, the total return for the year equals the return from the capital appreciation. The total return is calculated as follows:

$$\begin{aligned} R_T &= R_{CA} + RI = \frac{P_1 - P_0 + CF_1}{P_0} \\ &= \frac{\$44.00 - \$32.05 + \$0.00}{\$32.05} \\ &= 0.373, \text{ or } 37.3\% \end{aligned}$$

Expected Returns

- An **expected return** is the weighted average of possible investment returns, i.e. the sum of each possible return multiplied by the probability that it will occur

Equation 7.2

$$E(R_{\text{Asset}}) = \sum_{i=1}^n (p_i \times R_i)$$

Expected Return Example

- There is a 30% chance the total return on a stock will be -5.68% , a 30% chance it will be $+6.82\%$, a 30% chance it will be $+9.66\%$, and a 10% chance it will be $+13.64\%$. Calculate the expected return for the stock

$$\begin{aligned} E(R_{\text{Stock}}) &= [(0.30 \times -0.0568) + (0.30 \times 0.0682) + (0.30 \times 0.0966) + (0.10 \times 0.1364)] \\ &= -0.01704 + 0.02046 + 0.02898 + 0.01364 \\ &= 0.0460, \text{ or } 4.60\% \end{aligned}$$

Cost of Capital

Companies fund their activities using external and/or internal capital

1. External Capital

- i. Ordinary Shares
- ii. Preference Shares
- iii. Loan/Debt Capital

2. Internal Capital

- i. Retained Earnings

Cost of Capital

The **cost of capital** is the rate of return that the enterprise **must pay** to satisfy **the providers of funds**, and it reflects the riskiness of providing funds. It is ;

- a. The cost of funds that companies raise and use to fund their activities
- b. The return that the providers of funds expect to be paid

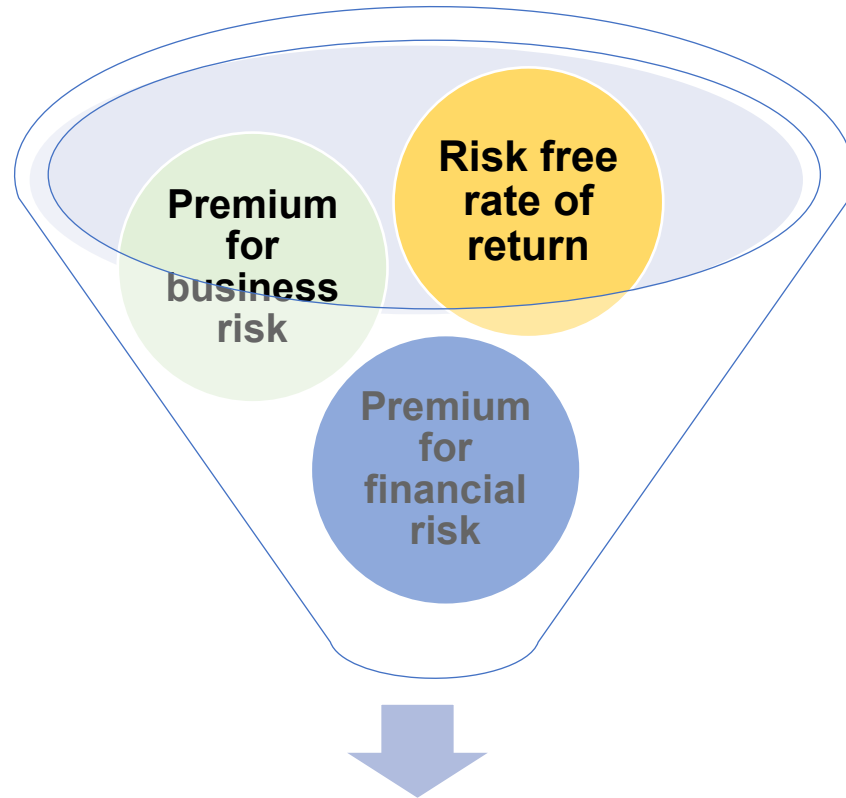
Therefore, it is the **minimum return** that a company should make on its own investments, to earn the cash flows out of which investors can be paid their return.

Cost of Capital

It is an **opportunity cost** of finance, because it is the *minimum return* that investors require. If they do not get this return, they will transfer some or all of their investment somewhere else.

- **Interest rate charged is the opportunity cost of lending for the bank** ; it can get as good a return from lending the money to someone else
- **The yield on shares is the opportunity cost to the shareholders of not investing somewhere else**

Cost of Capital



Cost Of Capital

What is the cheapest type of capital?

☐ **Risk-free rate of return**

return which would be required from an investment if it were completely free from risk

☐ **Premium for business risk**

required rate of return due to the existence of uncertainty about the future and about a firm's business prospects

☐ **Premium for financial risk**

required rate of return due to the danger of default because of high debt levels

The Cost of Equity

- ❑ Capital from Equity investors is obtained from new share issues or retained earnings
 - *Shareholders will **not provide funds** for a **new issue** of **shares** unless the return on their investment is sufficiently attractive.*
 - *Retained earnings also have a cost. This is an **opportunity cost**, the dividend forgone by shareholders.*

- ❑ The cost of equity can be estimated using
 - a. **The Dividend Valuation Model**
 - b. **The Dividend Growth Model**

The Cost of Equity: The Dividend Valuation Model (1)

Assumption

- i. Market value of shares is directly related to expected future dividends on the shares
- ii. Share issue (transaction) costs is zero or negligible
- iii. Ignores taxes

$$P_0 = \frac{d}{(1+k_e)} + \frac{d}{(1+k_e)^2} + \frac{d}{(1+k_e)^3} + \dots = \frac{d}{k_e}, \text{ so } k_e = \frac{d}{P_0}$$

Where k_e is the cost of equity capital

d is the annual dividend per share, starting at year 1 and then continuing annually in perpetuity.

P_0 is the ex-dividend share price (the price of a share where the share's new owner is **not** entitled to the dividend that is soon to be paid).

The Cost of Equity: The Dividend Valuation Model (2)

Example 1

Wrexham Plc has a dividend cover ratio of 4.0 times and expects zero growth in dividends. The company has one million £1 ordinary shares in issue and the market capitalisation (value) of the company is £50 million. After-tax profits for next year are expected to be £20 million.

What is the cost of equity capital?

Total dividends = 20 million/4 = £5 million.

K_e = 5/50 = **10%**.

The Cost of Equity: The Dividend Valuation Model (3)

Example 2

Wrexham Plc preference share has a 5% dividend rate, £100 stated value, and a price of £85.

What is the cost of this preferred stock?

$$D_{ps} = £100(0.05) = £5$$

$$K_{ps} = £5 / £85 = 0.0588, \text{ or } \mathbf{5.88\%}$$

The Cost of Equity: The Dividend Growth Model (1)

Assumption

- i. Shareholders will expect dividends to grow and not remain constant perpetually.
- ii. Dividends will grow at a constant rate
- iii. The required rate of return is known and does not change
- iv. Ignores taxes

Where,

$$P_0 = \frac{d_0(1+g)}{(K_e - g)} = \frac{d_1}{(K_e - g)}$$

- P_0 is the current market price (ex div)
 - d_0 is the current net dividend
 - K_e is the cost of equity capital
 - g is the expected annual growth in dividend payments
- and both K_e and g are expressed as proportions.

The Cost of Equity: The Dividend Growth Model (2)

Example

Wrexham Plc share has a current market value of 96p, and the last dividend was 12p. If the expected annual growth rate of dividends is 4%, **calculate the cost of equity capital.**

$$\begin{aligned}\text{Cost of equity capital} &= \frac{12(1+0.04)}{96} + 0.04 \\ &= 0.13 + 0.04 \\ &= 0.17 \\ &= \mathbf{17\%}\end{aligned}$$

Cost of Debt Capital (1)

The **cost of debt** is the return an enterprise must pay to its lenders.

The cost of debt capital represents:

1. The cost of **continuing to use the finance** rather than redeem the securities at their current market price.
2. The cost of raising **additional fixed interest capital** if we assume that the cost of the additional capital would be equal to the cost of that already issued.

Cost of Debt Capital (2)

Cost of irredeemable debt capital, paying interest i in perpetuity, and having a current ex-interest price P_o :

$$K_d = i / P_o$$

Where, K_d = cost of debt

i = interest rate

P_o = price of debt

Example

Wrexham Plc has issued bonds of £100 nominal value with annual interest of 9% per year, based on the nominal value. The current market price of the bonds is £90. **What is the cost of the bonds?**

$$K_d = 9/90 = 10\%$$

Cost of Debt Capital and Taxes (1)

- ❑ The interest on debt capital is likely to be an **allowable deduction** for purposes of **taxation** and so;
- ❑ the cost of debt capital and the cost of share capital are not properly comparable costs.
- ❑ This **tax relief** on interest ought to be recognised in computations.
- ❑ The after-tax cost of interest payments equals the pre-tax cost times 1 minus the tax rate:

The after-tax cost of irredeemable debt capital is:

$$k_{\text{Debt after-tax}} = k_{\text{Debt pretax}} \times (1 - t)$$

Cost of Debt Capital and Taxes (2)

Example

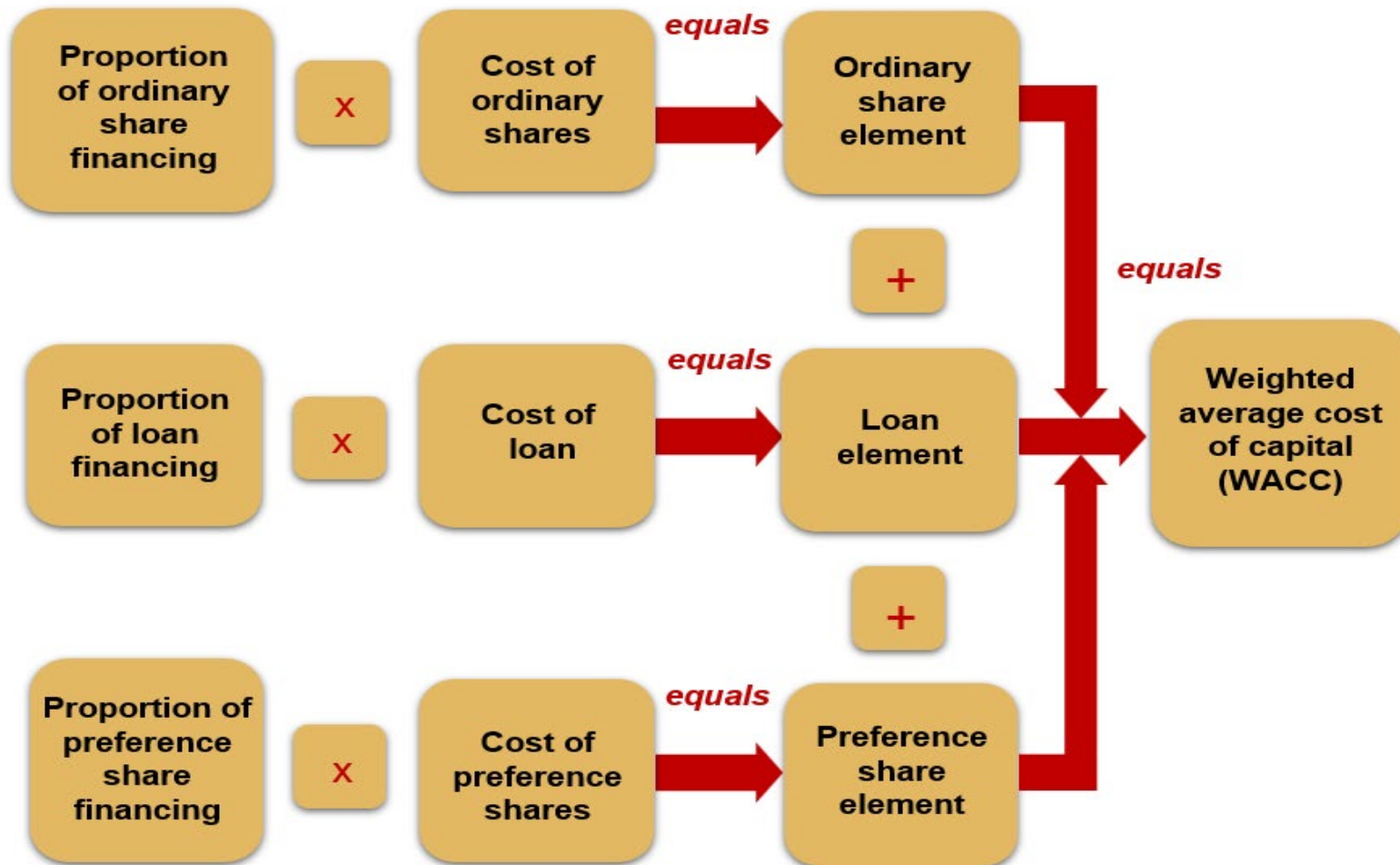
The before-tax cost of debt for a firm is 6% and the marginal tax rate is 20%. What is the after-tax cost of debt for the firm?

$$\begin{aligned}k_{\text{Debt after-tax}} &= 0.06 \times (1 - 0.20) \\ &= 0.048, \text{ or } 4.8\%\end{aligned}$$

The Weighted Average Cost of Capital (WACC)

- ❑ The **weighted average cost of capital** is calculated by weighting the costs of the individual sources of finance according to their **relative** importance as sources of finance.
- ❑ **Weighted average cost of capital** is the average cost of the company's finance (equity, bonds, bank loans) weighted according to the proportion each element bears to the total pool of capital.
 - a. WACC is based on market value, not book value
 - b. WACC is based on expected returns and is a forecast of the discount rate to apply to future cash flows
 - c. WACC assume the company's current capital structure will persist

Calculating WACC (1)



Calculating WACC (2)

If we divide the costs of capital into debt and equity portions of the firm, then we can use the above to arrive at the weighted average cost of capital (WACC) for a firm with only debt and equity:

$$k_{\text{Firm}} = x_{\text{Debt}} k_{\text{Debt}} + x_{\text{Equity}} k_{\text{Equity}}$$

Where X = proportion; and K = cost

Calculating WACC (3)

Example

Wrexham Plc total value is £4,000,000 and it has £300,000 debt. The cost of debt is 6% and the cost of equity is 10%.

What is the weighted-average cost of capital (WACC)?

$$x_d = £300,000 / £4,000,000 = 0.075 = 7.5\%$$

$$x_e = £3,700,000 / £4,000,000 = 0.925 = 92.5\%$$

$$k_{\text{Firm}} = (0.075)(0.06) + (0.925)(0.10) = 0.097, \text{ or } 9.7\%$$

Calculating WACC (4)

The after-tax version of the formula for the weighted-average cost of capital is:

$$\text{WACC} = x_{\text{Debt}} k_{\text{Debt pretax}} (1 - t) + x_{\text{ps}} k_{\text{ps}} + x_{\text{cs}} k_{\text{cs}}$$

Class Exercise

Calculate Wrexham Plc's WACC (previous example) if the tax rate applicable to debt capital is 20%?

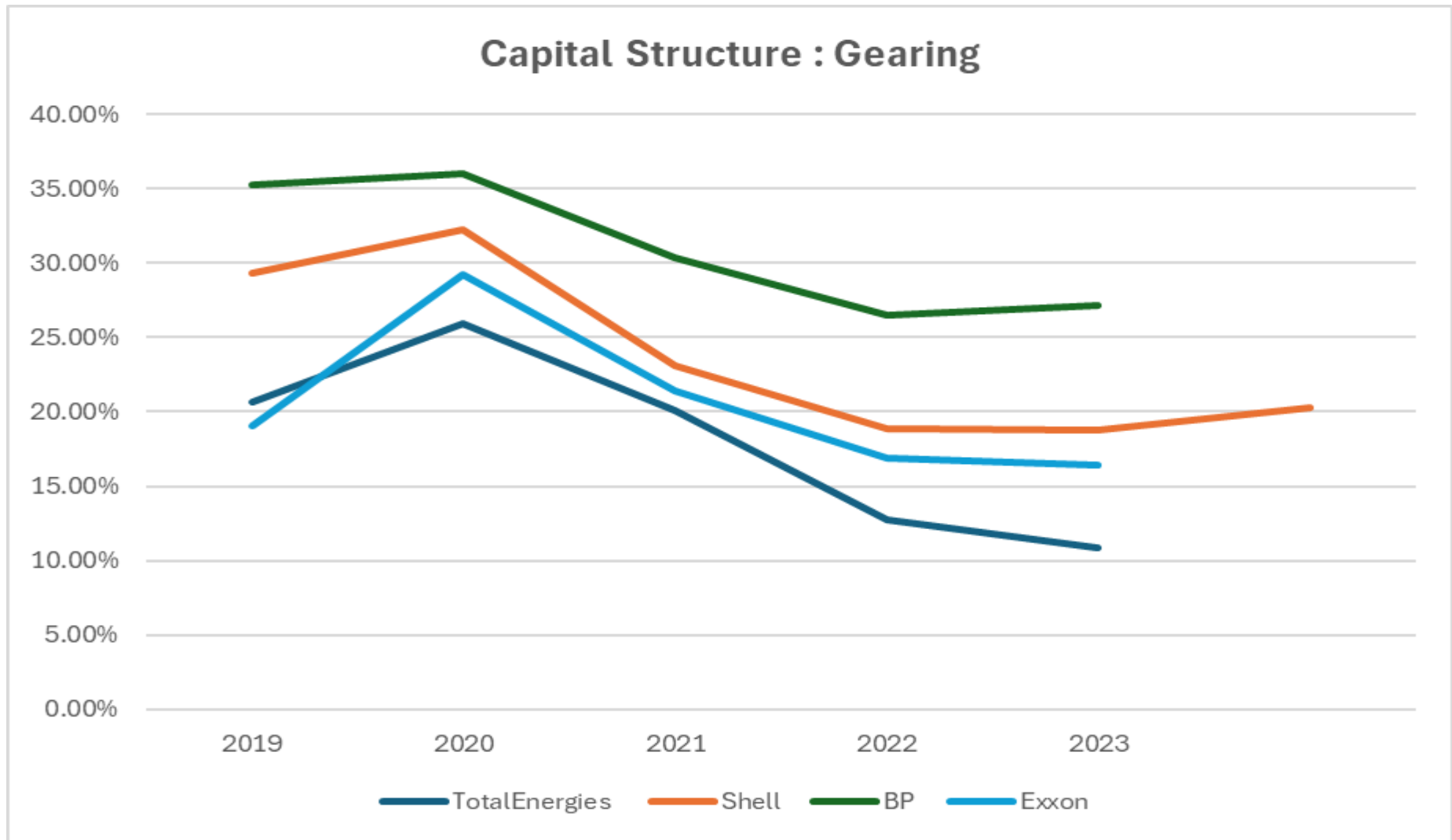
Limitations of WACC (1)

- i. Some elements of the WACC formula, like the cost of equity, are **not consistent values** and can be **reported differently** by different parties
- ii. WACC can be **difficult to calculate** for companies with complex balance sheets, such as those with **multiple types of debt and different interest rates**
- iii. WACC has many inputs, like interest rates and tax rates, which can be affected by market and economic conditions.
- iv. While valuation is forward looking, many of the inputs to WACC are **based on historical data**.

Limitations of WACC (2)

- v. WACC **relies on market values**, which can **fluctuate significantly**, leading to a WACC that may not accurately reflect long-term costs or risks
- vi. WACC assumes that the current capital structure **will not change** in the future
- vii. WACC may not be ideal for companies with **significantly different project risks** compared to their overall business risk.

Capital Structure: Oil Majors



Adapted from SEC 20-F fillings

Thank you, very much!

Any questions?



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